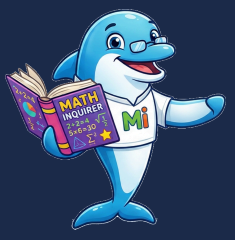
TUTOR GUIDE • LEVEL 1 • TERM 2 • LESSON 3 OF 10

Doubles and Near Doubles



Big idea: Double, then adjust by one. A double is two of the same number added. If two numbers are next to each other, use a double and then add 1 or take away 1.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Addition and Subtraction (Mental addition within 20 using doubles (1 + 1 to 10 + 10); Adding tens numbers within 100 (30 + 30); Near doubles as a mental strategy (7 + 8 is double 7 and 1 more); Adding more than two numbers (Go further only)) Strand label for the tutor only. FLAGGED NEW TOPIC: the plan marks this lesson as a new topic that needs an owner or tutor syllabus check before it is used with children (not yet done). Doubles to 10 + 10, tens doubles (30 + 30) and 'a double and 1 more' sit within the school Primary 1 idea of adding within 20 and 100. Beyond Primary 1 on purpose and only at Go further: doubling a two-digit number that is not a tens number (double 35, which needs renaming) and adding five numbers that are almost the same. Say this plainly to families. Do not describe the lesson as MOE-aligned or syllabus-aligned. Every Start here item is reached with ten-frames, dominoes or tens bars and counters after Explore.
Objectives	• Say and write the double of any number from 1 to 10 (4 + 4 = 8, double 7 is 14) using counters, ten-frames or a domino, and the double of a tens number (3 tens and 3 tens make 6 tens). • Add two numbers that are next to each other (7 + 8, 8 + 9) by using a double and then 1 more or 1 less, and say why the answer is 1 away from the double. • Choose which double to use for a near double (7 + 8 is double 7 and 1 more, or double 8 and 1 less) and spot when two numbers are not next to each other. • Go further: add numbers that are almost the same (8 + 9 + 8 + 7 + 8) by thinking of the number in the middle.
Joining cold	**Remember warm-up covers:** How many more make 10 on a ten-frame, shown with fingers (7 needs 3); Tens add: 3 tens and 3 tens make 6 tens (counted with tens bars); Same amount on both sides: 5 + 3 = 4 + 4 (move 1 counter from one plate to the other). Only counting to 20 and ten ones making a ten. Nothing from an earlier lesson is needed: the warm-up builds a ten-frame, tens bars and two plates with counters, and the double idea itself is taught in Explore with objects. Lesson 3 of Term 2 is fairly safe to miss: it stands alone after the warm-up. A child who misses it watches the recording and does workbook pages 1 and 2 before Lesson 4. Doubles help later (Lesson 6 uses the 2 table), but nothing in Lessons 4 or 5 needs them.
Child materials	• The printed workbook and the kit page (page 6), with the 20 counters (10 red, 10 yellow) cut out by a grown-up beforehand • If no printer is available: about 20 small things of two kinds (for example 10 buttons and 10 coins) • A pencil, a rubber and spare paper • Optional: a clean empty egg carton (a double of 6) and a set of dominoes
Tutor materials	• A slide deck and a paper copy of the kit page with red and yellow counters to show on camera if you wish • A visible clock No ruler is needed. Nothing needs to be bought: everything can be found at home. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears.

Tiers	Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder only: double and adjust by one (near doubles); find the middle number of several almost-equal numbers. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, using a double each time. Check the Go further items (double 35 and the five-number sum) against the key.
- Tell parents a week before: print the workbook pages and the kit page, cut out the kit page and find the things listed. Small things stay away from mouths and younger children.
- Have the slides open and check the kit page is visible on screen share. Have a paper copy of the kit page and counters to show on camera if you wish.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Start the recording after the greeting, and avoid first names while it runs.
- Be plain with families: this lesson is flagged as a new topic and still needs the owner or a tutor syllabus check. Doubles to 20, tens doubles and near doubles sit inside the school Primary 1 idea of adding within 20 and 100. Go further goes beyond school on purpose (double 35, five numbers that are almost the same). Do not call it MOE-aligned.

Safeguarding and recording (this lesson)

- Counters and small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. Keep small things away from younger brothers and sisters.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a board up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. A child may stay seated and move only their arms.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Doubles	1:28
2–8	**Remember warm-up**	6	3: Warm-up: fill the ten-frame, Tens and tens, The same on both sides	4:51
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:04
12–27	**Explore**	15	5: Explore: build a double, Bigger doubles, Bigger doubles, One more, one less, What did we notice?	12:34
27–39	**Mini-lesson**	12	10: Doubles to 10 + 10, Double a tens number, A near double: 7 + 8, A near double: 7 + 8, Which double do we use?, Next to each other?, Next to each other?, Your turn: add, Check: add, Almost the same: 8 + 9 + 7	10:07

39–43	**Brain break**	4	1: Stomp the double	3:18
43–71	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	22:59
71–77	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:54
77–83	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:58
83–86	**Look-back**	3	1: Look-back	2:23
86–90	**Flex time and goodbye**	4	1: Well done, inquirers!	0:46
	Total	**90**		**71:22**

The script needs 71:22 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 84:34. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 43 (start of workbook), minute 71 (talk time). If more than 2 minutes behind at minute 43, skip in this order: Start with: drop the Mini-lesson slide 'Almost the same' (only Go further needs it); Then: shorten Explore round 3 to one build (7 + 8); Then: shorten the Mini-lesson check slide to two questions; Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:28 of 2:00

0:00	SLIDE	Show slide **Doubles** (s-title)
0:15	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your kit page and counters. Pause or rewind any time.*
0:26	DO	Greet each child by first name as they arrive. Then start the recording. While it runs, avoid names. Check that you can see each child. [0:12]
0:38	SAY	Hello, inquirers! Please stay muted unless I invite you to speak. Cameras on if you can, so I can see you; cameras off is fine too. Pencil ready, and your counters, the kit page and your workbook on the table.
0:58	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is unmuted. You may mute and keep your camera off.
1:16	SAY	Today we learn about doubles: two of the same number. Then we use a double to add numbers that are next to each other.
	NOTE	Say the recording notice in your own words after the recording has started: the recording has the tutor, the shared screen and any unmuted voice; children may mute and keep the camera off. The routine and the four inquirer rules were introduced earlier in the term: do not re-teach them.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. The first recording of the term shows the full routine.

2:00 – 8:00 Remember warm-up

script 4:51 of 6:00

2:00	SLIDE	Show slide **Warm-up: fill the ten-frame** (s-rem1)
2:25	REC	*Recording: join in! Show your answers on your fingers before I say them. Pause if you need longer.*
2:34	SAY	Fingers and a ten-frame. A ten-frame has ten spaces. I say how many counters are in the frame. You show me, on your fingers, how many more fill it up.
2:49	ASK	Seven counters. How many more to make ten? [wait about 12 s for answers]
3:05	ASK	Four counters. How many more to make ten? [wait about 12 s for answers]
3:21	ASK	Nine counters. How many more? [wait about 12 s for answers]
3:36	ASK	Six counters. How many more? [wait about 12 s for answers]
3:50	SLIDE	Show slide **Tens and tens** (s-rem2)
4:15	SAY	Now tens. A tens bar is a ten. Three tens and three tens. Count the bars, then say the number.
4:25	ASK	Three tens and three tens. How many tens, and what is the number? [wait about 14 s for answers]
4:46	ASK	Two tens and two tens. What number? [wait about 12 s for answers]
5:01	ASK	Four tens and four tens. What number? [wait about 12 s for answers]
5:16	SLIDE	Show slide **The same on both sides** (s-rem3)
5:42	SAY	Last one. Two plates. Move one counter from the left plate to the right plate. Nothing is added and nothing is lost, so both sides stay equal. Five and three is the same as four and four.
6:00	ASK	Six and two is the same as five and what? [wait about 12 s for answers]
6:17	ASK	Seven and one is the same as six and what? [wait about 12 s for answers]

6:34	**ASK**	Five and three is the same as four and what? [wait about 12 s for answers]
	IF STUCK	If a child hesitates on how many more make ten, count the empty spaces in the ten-frame slowly with them. For tens, show the bars on the slide and count them in tens: ten, twenty, thirty. For the plates, move a counter on screen or on the child's own kit and count each side again.
	NOTE	Answers: 3, 6, 1, 4 (and 2 if you ask eight), then 6 tens (60), 40, 80, then 3, 2, 4. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20 and ten ones making a ten; the pictures do the rest.

8:00 – 12:00 Sneak-peek puzzles

script 3:04 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:47	**READ**	Puzzle one. Mei puts 4 red counters in one frame and 4 yellow counters in another frame. How many counters are there altogether? You may draw to help.
9:02	**CHILDREN**	Children count the counters and write the number on their own paper. [0:40]
9:42	**READ**	Puzzle two. Ben knows that 6 plus 6 is 12. He says that 6 plus 7 is 1 more than 12. Is Ben right? Circle Yes or No. Then write or draw how you know.
10:01	**CHILDREN**	Children write or draw their answer. [0:55]
10:56	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 8 counters. Puzzle 2 answer: yes; 6 + 7 is the double 6 + 6 with one more counter, so 13. Look for good ideas to praise later (a child who said it is the same and one more). Do not write scores or keep a list.

12:00 – 27:00 Explore

script 12:34 of 15:00

12:00	**SLIDE**	Show slide **Explore: build a double** (s-explore1)
12:25	**REC**	*Recording: pause now. Build the doubles with red and yellow counters, then try one more and one less. Press play when you have tried.*
12:37	**SAY**	Today your hands are the teachers. Put the ten-frame page in front of you. Take red counters and yellow counters. Put 3 red counters in one frame and 3 yellow counters in the next frame.
12:54	**ASK**	How many counters altogether? [wait about 14 s for answers]
13:10	**SAY**	Two of the same number. We call that a double. Double 3 is 6. Say it with me: three and three is six.
13:22	**SAY**	Now build double 4. Four red, four yellow.
13:26	**CHILDREN**	Children build 4 red and 4 yellow and count 8. [0:35]
14:01	**ASK**	What is double 4? [wait about 10 s for answers]
14:13	**SAY**	Double 5.
14:14	**CHILDREN**	Children build 5 red and 5 yellow: one frame is full. [0:35]
14:49	**ASK**	What is double 5? What do you notice about the red frame? [wait about 20 s for answers]
15:15	**SLIDE**	Show slide **Bigger doubles** (s-explore2)
15:40	**SAY**	Round two. Bigger doubles. Fill the first frame with red counters. Put the same number of yellow counters in the other frame. Then fill the first frame up to ten and say what is left.
15:58	**CHILDREN**	Build double 6: 6 red, 6 yellow. Move 4 yellow into the first frame: ten and 2. [0:45]
16:42	**ASK**	Double 6 is ten and what? [wait about 10 s for answers]
16:56	**CHILDREN**	Build double 7: ten and 4. [0:45]

17:40	**ASK**	Double 7 is ten and what? What is the number? [wait about 12 s for answers]
17:58	**CHILDREN**	Build double 8: ten and 6. [0:40]
18:38	**CHILDREN**	Build double 9: ten and 8. [0:30]
19:08	**ASK**	Double 8 and double 9. What are the numbers? [wait about 15 s for answers]
19:27	**SLIDE**	Show slide **Bigger doubles** (s-explore2-ans)
19:52	**SLIDE**	Show slide **One more, one less** (s-explore3)
20:17	**SAY**	Round three. Build double 7 again: seven red, seven yellow. Now put one more yellow counter in the yellow frame.
20:27	**CHILDREN**	Children add 1 yellow counter: 7 red and 8 yellow. [0:25]
20:52	**ASK**	Seven and eight. How many now? How do you know? [wait about 20 s for answers]
21:17	**SAY**	Now take two yellow counters away from the yellow frame. That leaves six yellow.
21:24	**CHILDREN**	Children take away 2 yellow counters: 7 red and 6 yellow. [0:25]
21:49	**ASK**	Seven and six. How many? Is that more or less than double 7? [wait about 20 s for answers]
22:16	**SAY**	Your turn. Eight red and nine yellow.
22:19	**CHILDREN**	Children build 8 and 9 (double 8, then 1 more): 17. [0:40]
22:59	**SLIDE**	Show slide **What did we notice?** (s-explore4)
23:24	**ASK**	What did you notice? What happens to the total when the yellow counters go up by 1? What happens when they go down by 1? [wait about 45 s for answers]
24:22	**SAY**	A double is easy to know. If the numbers are next to each other, find the double, then move one. That is the whole trick.
	IF STUCK	A child who adds all the counters again by counting from 1: ask them to start from the red frame, which they already know is a double, and add on one yellow counter. A child who has few counters: use two kinds of small things from the table, or kit squares and pencil dots.
	NOTE	Main room only: no breakout rooms. Children say their totals aloud; you cannot see builds on small tiles. Answers: double 3 is 6, double 4 is 8, double 5 is 10; double 6 is 12 (10 and 2), double 7 is 14 (10 and 4), double 8 is 16, double 9 is 18; 7 + 8 is 15, 7 + 6 is 13, 8 + 9 is 17. Say the word double once as a name and use it the same way every time. Do not teach 'near double' as a rule yet: let children say it.

27:00 – 39:00 Mini-lesson

script 10:07 of 12:00

27:00	**SLIDE**	Show slide **Doubles to 10 + 10** (s-m1)
27:25	**REC**	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
27:32	**SAY**	Look at the doubles we built. Double 1 is 2, double 2 is 4. The doubles go up by 2 each time. Read them with me up to double 10, which is 20.
27:48	**ASK**	Which doubles do you already know without counting? [wait about 20 s for answers]
28:12	**SLIDE**	Show slide **Double a tens number** (s-m2)
28:37	**SAY**	Doubles with tens. Three tens and three tens make six tens. Thirty and thirty is sixty. A double of a tens number works the same way.
28:50	**ASK**	Double 2 tens? Double 4 tens? What are the numbers? [wait about 20 s for answers]
29:15	**SLIDE**	Show slide **A near double: 7 + 8** (s-m3)
29:40	**SAY**	Now seven and eight. The numbers are next to each other. Look at the red frame. It is a double.
29:50	**ASK**	What is double 7? Then what is 7 plus 8? [wait about 25 s for answers]
30:20	**SLIDE**	Show slide **A near double: 7 + 8** (s-m3-ans)
30:45	**SLIDE**	Show slide **Which double do we use?** (s-m3b)

31:10	**SAY**	Two ways to see seven and eight. Use double 7 and add 1: the ringed counter. Or use double 8 and take 1 away: the dashed counter. Look at both pictures. How could we check the answer a different way?
31:30	**ASK**	Which double would you choose? Does the answer change? [wait about 25 s for answers]
32:00	**SLIDE**	Show slide **Next to each other?** (s-m3c)
32:24	**SAY**	Not every sum is a near double. Look at the two numbers. Are they next to each other? Count from the smaller number to the larger one: is it one step?
32:40	**ASK**	Six and seven. Are they next to each other? [wait about 12 s for answers]
32:56	**ASK**	Five and eight. Are they next to each other? How many steps from 5 to 8? [wait about 18 s for answers]
33:22	**ASK**	Nine and ten? [wait about 10 s for answers]
33:34	**SLIDE**	Show slide **Next to each other?** (s-m3c-ans)
33:59	**SLIDE**	Show slide **Your turn: add** (s-m4)
34:24	**READ**	Your turn. Add. Six plus seven. Eight plus nine. Four plus five. Nine plus ten.
34:32	**CHILDREN**	Children write four answers. [0:50]
35:22	**SLIDE**	Show slide **Check: add** (s-m4a)
35:47	**READ**	Thirteen, seventeen, nine, nineteen. Tick your own answers with a tick.
35:53	**SLIDE**	Show slide **Almost the same: 8 + 9 + 7** (s-m5)
36:18	**SAY**	One more trick. Eight, nine and seven are almost the same. The nine is one more than eight. The seven is one less than eight. They make up for each other, so it is like three eights.
36:37	**ASK**	What is three eights? Which number is in the middle? [wait about 25 s for answers]
	NOTE	Key words to use every time: double, one more, one less, next to each other. A double is two of the same number. A near double is two numbers that are next to each other. Do not say double and add 2: that is a different sum. Do not teach written column addition. Do not teach halving.
	IF STUCK	If a child gets 7 + 8 = 16, ask: which double did you use, and how many more or fewer is the second number? Count the extra yellow counter on the ten-frame picture. If a child adds 1 when they should take 1 away (8 + 9 from double 9), ask which number is the larger one.

39:00 – 43:00 Brain break

script 3:18 of 4:00

39:00	**SLIDE**	Show slide **Stomp the double** (s-break)
39:25	**REC**	*Recording: stand up and join in! Stomp and clap each double with me.*
39:32	**SAY**	Stand up and get a sip of water. I say a number. You stomp that many times with your left foot and the same with your right foot, then you say the double.
39:48	**CHILDREN**	Call 2, 3, 4, 5, 6. Children stomp (or tap, in a small space) both feet and say the double aloud together. [2:30]

43:00 – 71:00 Workbook

script 22:59 of 28:00

43:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
43:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
43:34	**READ**	Page 1, Start here. Question 1: look at the dominoes and fill in the missing numbers. Question 2: Mei has 6 red counters and 6 yellow counters. Fill in the missing numbers. Question 3: fill in the double of 2, 5, 7 and 9.
43:58	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [6:40]
50:38	**SLIDE**	Show slide **Check page 1** (s-ws1ans)

51:03	**READ**	Question 1: six, eight, ten. Question 2: twelve, twelve. Question 3: four, ten, fourteen, eighteen. Tick your own answers with a tick.
51:15	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
51:24	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
51:49	**READ**	Page 2. Question 4: look at the bars and fill in the missing numbers. Question 5: Ali has 8 stickers and Ben has 8 stickers. How many stickers do they have altogether? Question 6: tick True or False for each statement.
52:11	**CHILDREN**	Page 2, questions 4 to 6. [7:00]
59:11	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
59:36	**READ**	Question 4: six, sixty, eighty. Question 5: eight plus eight is sixteen; they have sixteen stickers. Question 6: true, false, true, true.
59:48	**SLIDE**	Show slide **Choose your next page** (s-choice)
60:13	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
60:29	**MOVE**	Stand and stretch for 30 seconds, then sit down and choose.
60:59	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [5:00]
	IF STUCK	Q1: count the dots on one half, then the other half. Q2: count the red counters, then the yellow counters; the two frames match. Q3: use the kit counters: build the double. Q4: count the bars on one side, then say how many tens in all. Q5: draw 8 and 8, or use counters. Q6: build each double and check.
	IF STUCK	A child who is still stuck ticks 'I am still stuck' on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder is for near doubles, which are new in this setting. Go further is mastery and extension; it goes past the school Primary 1 range on purpose (doubling 35, adding five numbers). Do not announce who is on which page. Slow class: the workbook part needs about 31 minutes against 28, so Start here may fill most of it. Take the extra minutes from the flex time, do not skip the answer slides, and say that pages 3 to 5 are for home.

71:00 – 77:00 Talk time

script 4:54 of 6:00

71:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
71:25	**REC**	*Recording: pause. Tell a grown-up who is right, and show them with counters.*
71:32	**READ**	Mei adds 7 plus 8. She says it is 1 more than double 7. Ali says it is 1 less than double 8. Is Mei right? Is Ali right? Use counters to show.
71:50	**CHILDREN**	Build double 7 and double 8 on your frames. What happens when you move one? [0:50]
72:40	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:50]
73:30	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 80 s for answers]
74:56	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
75:20	**SAY**	Remember puzzle two? Six and six is twelve. Six and seven has one more counter, so it is thirteen. Yes, Ben is right.
75:32	**ASK**	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 15 s for answers]

	NOTE	Model answer: both are right. Double 7 is 14 and 1 more is 15. Double 8 is 16 and 1 less is 15. Both give 15. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

77:00 – 83:00 Exit ticket

script 4:58 of 6:00

77:00	SLIDE	Show slide **Exit ticket** (s-exit)
77:25	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
77:33	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
77:41	READ	E1. Fill in the missing number. Double 9 is what? E2. Double 30 is what? E3. Add. 9 plus 8.
77:52	CHILDREN	Children answer E1 to E3 on their own. [3:30]
81:22	SLIDE	Show slide **Check the exit ticket** (s-exitans)
81:47	READ	Eighteen. Sixty. Seventeen.
81:49	SAY	Tick your own ticket with a tick. Well done. Put a tick by each one you got right.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child's result.

83:00 – 86:00 Look-back

script 2:23 of 3:00

83:00	SLIDE	Show slide **Look-back** (s-look)
83:25	REC	*Recording: pause and fill in your look-back box. You can pass.*
83:30	SAY	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick 'I am still stuck', ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
84:03	CHILDREN	Children fill in the look-back box on page 5. [1:20]

86:00 – 90:00 Flex time and goodbye

script 0:46 of 4:00

86:00	SLIDE	Show slide **Well done, inquirers!** (s-end)
86:25	SAY	Well done, inquirers. Next lesson is Odd plus Odd. Please have your next workbook with the kit page of number cards cut out, 20 counters or small things of one kind, a pencil and paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: what if?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: Mei says double 9 is the same as 10 and 8. Can you see why? Build it with your counters.
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: double 9 is 18; move 1 counter from one 9 to the other: 10 and 8. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 4-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Mei says: a near double is always 1 more than the double of the smaller number. Test it with 4 + 5, 9 + 10 and 12 + 13. Does it always work?	*Recording: pause. Test 4 + 5, 9 + 10 and 12 + 13. Is each total one more than the smaller double?*	Yes. 4 + 5 = 9 (double 4 is 8), 9 + 10 = 19 (double 9 is 18), 12 + 13 = 25 (double 12 is 24). The bigger number is always 1 more than the smaller, so the total is 1 more than its double.
Make a puzzle for your grown-up	4 (optional)	Write three near doubles, such as 6 + 7. Give them to your grown-up. Can they find each total with a double?	*Recording: pause. Write three near doubles for a grown-up and check their answers with counters.*	No single answer. Check that the two numbers are next to each other and that each total is right, for example 6 + 7 = 13, 8 + 9 = 17, 5 + 6 = 11.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never seen a double. Teach them how to add 7 + 8 with a double.	*Recording: pause. Teach a grown-up how to add 7 + 8 with a double.*	Double 7 is 14. The 8 is 1 more than 7, so add 1: 15. Or double 8 is 16, take 1 away: 15.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Mei puts 4 red counters in one frame and 4 yellow counters in another frame. How many counters are there altogether? You may draw to help.

Answer: 8

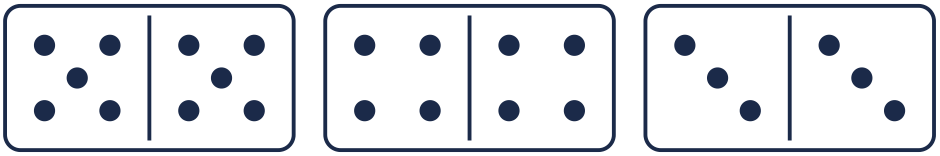
Puzzle 2 (think): Ben knows that 6 + 6 = 12. He says that 6 + 7 is 1 more than 12. Is Ben right? Circle Yes or No. Then write or draw how you know.

Answer: Yes. 6 + 7 is 6 + 6 with one more counter, so 12 + 1 = 13.

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Look at the dominoes. Fill in the missing numbers.**

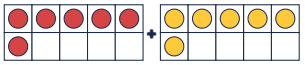


(a) 3 + 3 = ☐
(b) 4 + 4 = ☐
(c) 5 + 5 = ☐☐

Answer: 6, 8, 10

Hint (say this first): Count the dots on one half. The other half has the same number.
Worked solution: Each domino has two halves with the same number of dots. 3 + 3 = 6. 4 + 4 = 8. 5 + 5 = 10.
Common mistakes: Counting only one half of the domino.; Writing 5 + 5 = 55 (joining the digits).
Time for a middle child: about 1.5 min.

S2 **Mei has 6 red counters and 6 yellow counters. Fill in the missing numbers.**



(a) 6 + 6 = ☐☐
(b) Double 6 is ☐☐.

Answer: 6 + 6 = 12; double 6 is 12

Hint (say this first): Count the red counters, then count on the yellow counters. Or fill the first frame to ten.
Worked solution: The first frame has 6 red counters and the second has 6 yellow counters. 6 + 6 = 12, so double 6 is 12.
Common mistakes: Counting only one frame (6).; Writing 66.
Time for a middle child: about 2 min.

S3 **Fill in the missing numbers.**

(a) Double 2 is ☐.
(b) Double 5 is ☐☐.
(c) Double 7 is ☐☐.
(d) Double 9 is ☐☐.

Answer: double 2 is 4; double 5 is 10; double 7 is 14; double 9 is 18

Hint (say this first): Use your counters or the ten-frame page. A double is two of the same number.
Worked solution: Double 2 is 2 + 2 = 4. Double 5 is 5 + 5 = 10. Double 7 is 7 + 7 = 14. Double 9 is 9 + 9 = 18.
Common mistakes: Writing the number again (double 7 is 7).; Adding 2 to the number (double 7 is 9).
Time for a middle child: about 3 min.

S4 **Look at the bars. Fill in the missing numbers.**



(a) 3 tens + 3 tens = ☐ tens
(b) 30 + 30 = ☐☐
(c) Double 40 is ☐☐.

Answer: 3 tens + 3 tens = 6 tens; 30 + 30 = 60; double 40 is 80

Hint (say this first): Each bar is a ten. Count the bars on both sides.
Worked solution: 3 tens and 3 tens make 6 tens. 6 tens is 60. Double 4 tens is 8 tens, which is 80.
Common mistakes: Writing 6 for 30 + 30.; Writing 33 or 303.
Time for a middle child: about 2.5 min.

S5 **Ali has 8 stickers. Ben has 8 stickers. How many stickers do they have altogether?**



(a) ☐ + ☐ = ☐☐
(b) They have ___ stickers altogether.

Answer: 8 + 8 = 16; they have 16 stickers

Hint (say this first): They have the same number of stickers. This is a double.
Worked solution: Ali has 8 and Ben has 8. 8 + 8 = 16. They have 16 stickers altogether.
Common mistakes: Writing 8 (only one child).; Writing 88.
Time for a middle child: about 2.5 min.

True/False statements: Double 4 is 8.; Double 6 is 14.; 9 + 9 is a double.; Double 30 is 60.

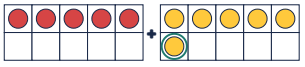
S6 **Read each statement. Decide if it is True or False.**

Answer: True, False, True, True

Hint (say this first): Work out each double first. Then decide if the statement matches.
Worked solution: Double 4 is 8: True. Double 6 is 12, not 14: False. 9 + 9 is two of the same number, so it is a double: True. Double 30 is 60: True.
Common mistakes: Marking Double 6 is 14 as true because 6 + 6 is close to 14.; Saying 9 + 9 is not a double because 9 is a big number.
Time for a middle child: about 2.5 min.

Think harder (page 3): the taught idea in a new setting

T1 **Look at the counters. The ringed counter is 1 more. Fill in the missing numbers.**



(a) Double 5 is ☐☐.
(b) 5 + 6 = ☐☐

Answer: double 5 is 10; 5 + 6 = 11

Hint (say this first): Count the red counters. The yellow counters without the ring match them.
Worked solution: 5 red counters and 5 yellow counters make double 5 = 10. The ringed yellow counter is 1 more, so 5 + 6 = 10 + 1 = 11.
Common mistakes: Writing 10 for both parts.; Adding 2 instead of 1.
Time for a middle child: about 3 min.

T2 **Add.**

(a) 3 + 4 = ☐
(b) 8 + 7 = ☐☐
(c) 10 + 11 = ☐☐
(d) 2 + 3 = ☐

Answer: 7, 15, 21, 5

Hint (say this first): Find the double of the smaller number. Then move one.
Worked solution: 3 + 4: double 3 is 6, and 1 more is 7. 8 + 7: double 7 is 14, and 1 more is 15. 10 + 11: double 10 is 20, and 1 more is 21. 2 + 3: double 2 is 4, and 1 more is 5.
Common mistakes: Taking 1 away instead of adding 1 after the smaller double.; Writing the double itself (6 for 3 + 4).
Time for a middle child: about 4 min.

T3 **Which sum is not a near double? Write the number in the bracket.**

Answer: (3) 5 + 8

Hint (say this first): A near double has two numbers that are next to each other. Look at each pair.
Worked solution: 6 and 7 are next to each other. 8 and 9 are next to each other. 9 and 10 are next to each other. 5 and 8 are not next to each other (they are 3 apart), so 5 + 8 is not a near double.
Common mistakes: Choosing 9 + 10 because 10 looks different.; Looking at the total instead of the two numbers.
Time for a middle child: about 3 min.

T4 **Kumar adds 8 + 9. Step 1: 8 + 8 = 16. Step 2: 16 - 1 = 15. Answer (a) and (b).**

(a) Is Kumar correct? Circle Yes or No.
(b) 8 + 9 = ☐☐

Answer: No; 8 + 9 = 17

Hint (say this first): The second number is greater than 8. Is the total more or less than the double?
Worked solution: 9 is 1 more than 8, so the total is 1 more than 16: 16 + 1 = 17. Kumar took 1 away instead of adding 1.
Common mistakes: Saying Yes because 16 - 1 looks like a good step.; Writing 15 for the right answer.
Time for a middle child: about 3 min.

Go further (page 4): mastery and extension

G1 **Add 8 + 9 + 8 + 7 + 8. Show your working on the lines.**

The total is ☐☐.

Answer: 40

Hint (say this first): Look at the 9 and the 7. How far is each from 8?
Worked solution: The 9 is 1 more than 8 and the 7 is 1 less than 8, so they make up for each other. The five numbers are the same as five 8s: 8 + 8 + 8 + 8 + 8 = 40. Another way: 8 + 8 = 16, 9 + 7 = 16, and 16 + 16 + 8 = 40.
Common mistakes: Adding from left to right and slipping.; Leaving one number out.
Time for a middle child: about 5 min.

G2 **Ben says 9 + 10 + 11 is the same as 10 + 10 + 10. Answer (a) and (b). Show your working on the lines.**

(a) Is Ben correct? Circle Yes or No.
(b) 19 + 20 + 21 = ☐☐

Answer: Yes; 19 + 20 + 21 = 60

Hint (say this first): The 9 is 1 less than 10. The 11 is 1 more than 10.
Worked solution: 9 is 1 less than 10 and 11 is 1 more than 10, so they make up for each other: 9 + 10 + 11 = 30 = 10 + 10 + 10. In the same way 19 + 20 + 21 is three 20s, which is 60.
Common mistakes: Saying No because the numbers look different.; Writing 50 for the three numbers.
Time for a middle child: about 5 min.

G3 **Find double 35. Fill in the missing numbers. Show your working on the lines.**

(a) Double 30 is ☐☐.
(b) Double 5 is ☐☐.
(c) Double 35 is ☐☐.

Answer: 60, 10, 70

Hint (say this first): Split 35 into 30 and 5. Double each part, then put them together.
Worked solution: 35 is 30 and 5. Double 30 is 60. Double 5 is 10. 60 + 10 = 70. So double 35 is 70.
Common mistakes: Writing 610 (joining 60 and 10).; Forgetting to double the 5.
Time for a middle child: about 5 min.

Investigation (page 5): Make two groups the same

Easy first step: Draw 6 dots and 8 dots. Move 1 dot from the 8 to the 6. What do you see?
Then: Draw dots for each pair. Try 5 and 9, then 7 and 8. Can you move dots so both groups are the same? When does it work? Write what you notice.

> **Model answer: It works for 6 and 8 (both 7), and for 5 and 9 (both 7): the two numbers are 2 apart or 4 apart and the total is a double (14). It does not work for 7 and 8: the total is 15 and one dot is always left over.**

Look for: A child who tries several pairs, moves dots, and notices that when the total is a double the two groups can be made the same, and that 7 and 8 always leave one dot over. Credit any clear reason. Do not use the words odd or even yet: that is the next lesson.
Common mistakes: Trying only one pair.; Moving too many dots across, so the groups swap instead of matching.

Talk and explain (minutes 71 to 77)

Mei adds 7 + 8. She says it is 1 more than double 7. Ali says it is 1 less than double 8. Is Mei right? Is Ali right? Use counters to show. Sentence starter: I think ... because ...

> **Model answer: Both are right. Double 7 is 14 and 1 more is 15. Double 8 is 16 and 1 less is 15. Both ways give 15, because 7 is one less than 8: 7 + 8 is double 7 and 1 more, or double 8 and 1 less.**

Look for: A reason about moving one counter: one more than the smaller double, or one less than the larger double. Not just both are 15. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 77 to 83): 3 questions

E1 **Fill in the missing number.**
Double 9 is ☐☐.

> **Answer: 18**

Worked solution: 9 + 9 = 18. Move 1 from one 9 to the other: 10 and 8.
Common mistakes: Writing 9 or 11.; Writing 99.

E2 **Fill in the missing number.**
Double 30 is ☐☐.

> **Answer: 60**

Worked solution: Double 3 is 6, so double 3 tens is 6 tens: 60.
Common mistakes: Writing 6.; Writing 33 or 303.

E3 **Add.**
9 + 8 = ☐☐

> **Answer: 17**

Worked solution: Double 8 is 16 and 1 more is 17. Or double 9 is 18 and 1 less is 17.
Common mistakes: Writing 18 (the double of 9).; Writing 16.

Look-back (minutes 83 to 86)

Children tick or write: *Something I did well*; *The hardest part was* (remembering the doubles; adding 1 more or 1 less; doubling tens like 30 + 30; choosing which double (Think harder); something else); *What helped me* (counters or ten-frames; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Fill in the missing numbers. (a) Double 4 is []. (b) Double 6 is [2]. (c) Double 8 is [2].	8, 12, 16
H2	Start here	Mei has 7 red marbles and 7 blue marbles. How many marbles does she have altogether? She has ___ marbles.	14
H3	Start here	Fill in the missing numbers. (a) Double 20 is [2]. (b) Double 40 is [2].	40, 80
H4	Think harder	Add. (a) 6 + 7 = [2] (b) 9 + 10 = [2]	13, 19
H5	Go further	Add 7 + 8 + 7 + 6 + 7. The total is ___.	35

Online practice (`out/l13/practice`, key `t2l13`): 14 items; 11 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to use doubles to add quickly and to say why a near double is 1 more or 1 less. This week: doubles to 10 + 10, tens doubles and near doubles.

Term line (plan, section 5): Your child is learning to draw a story, to use groups and sharing, and to test a claim with an example. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Doubles** s-title	TODAY WE WILL: / Find doubles, then use them to add.	Lesson 3 of Term 2, 90 minutes. Objective: know the doubles to 10 + 10 and double a tens number, then add numbers that are next to each other (7 + 8) with a double and 1 more or 1 less. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 27, mini-lesson 27 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Say the recording notice in one line: you and the screen only; a child may keep the camera off.
2	2–8	**Warm-up: fill the ten-frame** s-rem1	Show it on your fingers / [picture: ten_frame] / A ten-frame has 10 spaces. How many more counters make 10? / 7 4 9 6 8	A ten-frame has ten spaces. Count the empty spaces to see how many more make ten. Say each number; children show on their fingers how many more make ten. Answers: 7 needs 3, 4 needs 6, 9 needs 1, 6 needs 4, 8 needs 2. Hands up on camera is fine. No writing, no marks. A child who is new can count the empty spaces in the picture.
3	2–8	**Tens and tens** s-rem2	[picture: bundles 3 tens 0 ones] / 3 tens and 3 tens make ___ tens. That is the number ___. / 3 tens + 3 tens 2 tens + 2 tens 4 tens + 4 tens	A tens bar is a ten. Three bars and three more bars make six bars: sixty. Ask each pair: three tens and three tens (6 tens, 60), two tens and two tens (4 tens, 40), four tens and four tens (8 tens, 80). Say the tens word slowly. A child who has not met tens only needs to hear and say the number words; show the bars on the slide and count them in tens.
4	2–8	**The same on both sides** s-rem3	[picture: plates] / 5 + 3 / [picture: plates] / 4 + 4 / Move 1 counter across. The total stays the same. / 6 + 2 = 5 + ___ 7 + 1 = 6 + ___	Two plates. Moving one counter from the left plate to the right plate does not change how many counters there are, so 5 + 3 and 4 + 4 are the same total. Ask: 6 + 2 = 5 + 3 (the missing number is 3), 7 + 1 = 6 + 2 (2), then 5 + 3 = 4 + 4 (4). This is the equals-means-the-same idea that near doubles use. Show the plates picture if a child is unsure.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Mei puts 4 red counters in one frame and 4 yellow counters in another frame. / How many counters are there altogether? / You may draw to help. / [picture: ten_frame] / PUZZLE 2: think: / Ben knows that 6 + 6 = 12. / He says that 6 + 7 is 1 more than 12. / Is Ben right? Circle Yes or No. Then write or draw how you know.	Puzzle 1 (8 counters, 4 red and 4 yellow in two ten-frames) is an access question: any child can try it by counting or drawing. Puzzle 2 is a thinking question on the lesson's key idea: one more counter makes one more in the total. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera: children keep their papers and check their own thinking at Talk time. There is no private chat. Answers: 8; yes (6 + 7 is 6 + 6 and one more, so 13). Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.
6	12–27	**Explore: build a double** s-explore1	Lay the ten-frame page in front of you. / Put 3 red counters in one frame. / Put 3 yellow counters in the other frame. / How many altogether? Then try 4 and 4, and 5 and 5. / [picture: ten_frame] / Two of the same number is a double.	Round 1, about 4 minutes. Objects first, no writing. Children put 3 red counters in one frame and 3 yellow counters in the other, then count 6. Name the idea once: two of the same number is a double, so double 3 is 6. Then double 4 (8) and double 5 (10; the red frame is full). A child without the kit page uses two kinds of small things on a drawn ten-frame, or kit squares and pencil dots.

7	12–27	**Bigger doubles** s-explore2	Build double 6, 7, 8 and 9. Fill the first frame up to ten. How many are left? / 6 + 6 7 + 7 8 + 8 9 + 9 / I move ..., so it is ten and ...	Round 2, about 4 minutes, main room. Each child builds the double with red and yellow counters, then moves yellow counters into the first frame until it is full. The picture shows double 7: the first frame has 7 red and 3 yellow, and 4 yellow are left, so ten and 4. Answers: double 6 is 12 (ten and 2), double 7 is 14 (ten and 4), double 8 is 16 (ten and 6), double 9 is 18 (ten and 8). This links to making a ten. Do not name children. The picture of the answer is on the next slide, shown after the children have said it.
8	12–27	**Bigger doubles** s-explore2-ans	Build double 6, 7, 8 and 9. Fill the first frame up to ten. How many are left? / 6 + 6 7 + 7 8 + 8 9 + 9 / [picture: ten_frame] / I move ..., so it is ten and ...	Round 2, about 4 minutes, main room. Each child builds the double with red and yellow counters, then moves yellow counters into the first frame until it is full. The picture shows double 7: the first frame has 7 red and 3 yellow, and 4 yellow are left, so ten and 4. Answers: double 6 is 12 (ten and 2), double 7 is 14 (ten and 4), double 8 is 16 (ten and 6), double 9 is 18 (ten and 8). This links to making a ten. Do not name children. The picture of the answer is on the next slide, shown after the children have said it. (answer shown after the children have answered)
9	12–27	**One more, one less** s-explore3	[picture: ten_frame] / Double 7 / [picture: ten_frame] / 1 more yellow counter / Build 7 + 8, then 7 + 6. Then build 8 + 9.	Round 3, about 5 minutes. Children build double 7, add one yellow counter (7 + 8 = 15), then take two yellow away (7 + 6 = 13) so they see it is one less than the double. Then they build 8 + 9: double 8 is 16 and one more is 17. Ask: is it more or less than the double? How do you know? Children count the extra counter, not the whole frame again.
10	12–27	**What did we notice?** s-explore4	What is double 7? What is 7 + 8? / What happens when the yellow counters go up by 1? / What happens when they go down by 1?	Take one or two answers from volunteers. Key ideas: a double is two of the same number; when one number goes up by 1, the total goes up by 1; when it goes down by 1, the total goes down by 1; so for numbers that are next to each other, find the double and move one. Repeat a child's answer in your own words so the recording picks it up.
11	27–39	**Doubles to 10 + 10** s-m1	Double | Number sentence | Total / Double 1 | 1 + 1 | 2 / Double 2 | 2 + 2 | 4 / Double 3 | 3 + 3 | 6 / Double 4 | 4 + 4 | 8 / Double 5 | 5 + 5 | 10 / Double 6 to 10: 12, 14, 16, 18, 20	A summary of what children built. Read the table with them: double 1 is 2, double 2 is 4, double 3 is 6, double 4 is 8, double 5 is 10, and then 12, 14, 16, 18, 20 for double 6 to 10. The totals go up by 2 each time. Ask which doubles they already know without counting; most children know 5 + 5 and 3 + 3. Take volunteers' answers and repeat them for the recording.
12	27–39	**Double a tens number** s-m2	[picture: bundles 3 tens 0 ones] / 3 tens / [picture: bundles 3 tens 0 ones] / 3 tens / 3 tens + 3 tens = 6 tens. 30 + 30 = 60	Double 3 is 6, so double 3 tens is 6 tens: 30 + 30 = 60. A double of a tens number works the same way as the double of a one-digit number. Ask: double 2 tens (40) and double 4 tens (80). Say the tens word slowly: forty, eighty. Beyond this slide, doubling numbers like 35 is for Go further only.
13	27–39	**A near double: 7 + 8** s-m3	[picture: ten_frame] / The numbers are next to each other: 7 and 8.	Seven and eight are next to each other. Find the double: double 7 is 14. The yellow frame has one more counter, so add 1: 14 + 1 = 15. Note that 7 + 8 is not a double itself: the numbers are different. We call a sum like this a near double. Answer: 15. Ask first (what is double 7, and so what is 7 + 8?); the working line comes on the answer slide.

14	27–39	**A near double: 7 + 8** s-m3-ans	[picture: ten_frame] / The numbers are next to each other: 7 and 8. / 7 + 8 = 7 + 7 + 1 = 14 + 1 = 15	Seven and eight are next to each other. Find the double: double 7 is 14. The yellow frame has one more counter, so add 1: 14 + 1 = 15. Note that 7 + 8 is not a double itself: the numbers are different. We call a sum like this a near double. Answer: 15. Ask first (what is double 7, and so what is 7 + 8?); the working line comes on the answer slide. (answer shown after the children have answered)
15	27–39	**Which double do we use?** s-m3b	[picture: ten_frame] / Double 7 and 1 more / [picture: ten_frame] / Double 8 and 1 less / Both ways: 7 + 8 = 15	7 + 8 can be found two ways. Use double 7 (14) and add 1, or use double 8 (16) and take 1 away. Both give 15. Ask and listen: which would you choose? Does the answer change? Do not state a rule: some children prefer the smaller double and 1 more. Answer: 15 both ways. The ringed counter is the 1 more; the dashed counter is the 1 taken away.
16	27–39	**Next to each other?** s-m3c	6 + 7 5 + 8 9 + 10 / A near double has two numbers that are next to each other.	A near double has two numbers that are next to each other, one step apart. 6 and 7: yes. 5 and 8: no, it is three steps (5, 6, 7, 8), so it is not a near double. 9 and 10: yes. Ask each pair first; the table with the answers comes on the next slide. Let children count the steps on their fingers. This prepares question T3 on page 3.
17	27–39	**Next to each other?** s-m3c-ans	6 + 7 5 + 8 9 + 10 / Sum | Next to each other? / 6 + 7 | Yes: 6, 7 / 5 + 8 | No: 5, 6, 7, 8 / 9 + 10 | Yes: 9, 10 / A near double has two numbers that are next to each other.	A near double has two numbers that are next to each other, one step apart. 6 and 7: yes. 5 and 8: no, it is three steps (5, 6, 7, 8), so it is not a near double. 9 and 10: yes. Ask each pair first; the table with the answers comes on the next slide. Let children count the steps on their fingers. This prepares question T3 on page 3. (answer shown after the children have answered)
18	27–39	**Your turn: add** s-m4	Write on your paper. Use a double to help. / 6 + 7 8 + 9 4 + 5 9 + 10	Each child writes four answers on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group. Some children may count on: that is fine, ask them to try a double as well. Possible doubles: 6 + 7 is double 6 and 1 (13); 8 + 9 is double 8 and 1 (17); 4 + 5 is double 4 and 1 (9); 9 + 10 is double 9 and 1 (19).
19	27–39	**Check: add** s-m4a	Sum | How we found it / 6 + 7 | Double 6 is 12. 12 + 1 = 13 / 8 + 9 | Double 8 is 16. 16 + 1 = 17 / 4 + 5 | Double 4 is 8. 8 + 1 = 9 / 9 + 10 | Double 9 is 18. 18 + 1 = 19	Answers: 13, 17, 9, 19. Read each answer and the way we found it aloud so a child watching the recording can mark their own work. A child who used the larger double and took one away (for example 16 - 1 for 8 + 9 is wrong, but 18 - 1 is right) can say which double they used; any correct way is fine.
20	27–39	**Almost the same: 8 + 9 + 7** s-m5	8 9 7 / The 9 is 1 more than 8. The 7 is 1 less than 8. / 8 + 9 + 7 = 8 + 8 + 8 = 24	A worked example for Go further. The 9 is one more than 8 and the 7 is one less than 8, so they make up for each other: the sum is three 8s, which is 24. Check: 8 + 9 = 17 and 17 + 7 = 24. Keep it to about a minute and do not rush: ask which number is in the middle. Children who are not ready can skip this idea; it is only needed for Go further.
21	39–43	**Stomp the double** s-break	I say a number. Stomp softly that many times on each foot. / Then say the double. Try 2, 3, 4, 5, then 6. / Stand up. Sip some water.	Four minutes. Stand up on camera. Call 2, 3, 4, 5, 6 (answers 4, 6, 8, 10, 12). Children stomp that many times with the left foot, then the same with the right foot, softly, or tiptoe or tap their fingers in a small space, and say the double aloud together. No equipment.
22	43–71	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 3: children may use the ten-frame page and the red and yellow counters.

23	43–71	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children tick their own answers with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 6, 8, 10; Q2 12, 12; Q3 4, 10, 14, 18.
24	43–71	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Take your time. / questions S4, S5, S6	Read questions 4 to 6 aloud. Question 4 uses tens bars; question 5 is a story with two groups of 8; question 6 is a True or False table, so children tick one box on each line. Children may use their kit counters to decide.
25	43–71	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can mark their own work. Question 6 answers: Double 4 is 8 (true), double 6 is 14 (false: it is 12), 9 + 9 is a double (true), double 30 is 60 (true). Never ask a child to show a wrong answer to the group.
26	43–71	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Add numbers that are next to each other / Go further (page 4) | Add more numbers and bigger doubles / Investigation (page 5) | Make two groups the same / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
27	71–77	**Talk and explain** s-talk	MEI AND ALI ADD 7 + 8: / Mei says it is 1 more than double 7. / Ali says it is 1 less than double 8. / Is Mei right? Is Ali right? Use counters to show. / I think ... because ...	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: both are right. Double 7 is 14 and 14 + 1 = 15. Double 8 is 16 and 16 - 1 = 15. Both ways give 15 because one counter moves in or out of the double. Repeat each child's answer in your own words for the recording.
28	71–77	**Back to the sneak-peek** s-sneakans	[picture: ten_frame] / 6 + 6 / [picture: ten_frame] / 6 + 7 / Yes! One more counter makes the total 1 more.	Reveal puzzle 2 now. Say: many of you wrote a good reason. Do not read out anyone's earlier answer. The pictures show 6 + 6 = 12 and 6 + 7 = 13: Ben is right because the second frame has one more counter, so the total is one more.
29	77–83	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud (the questions are on the slide). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
30	77–83	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: 18, 60, 17.
31	83–86	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
32	86–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Odd + Odd: Always, Sometimes, Never / Have ready: your Lesson 4 workbook with its kit page of number cards cut out, 20 counters or small things of one kind, pencil and paper.	Say goodbye and thank the grown-ups. Minutes 86 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.

33	optional	**Go further: what if?** s-opt	Mei says double 9 is the same as 10 + 8. Can you see why? / [picture: ten_frame]	Optional, only if you are on time. Answer: double 9 is 18; move 1 counter from the second 9 into the first frame and the first frame is full (10) and 8 are left, so 10 + 8 = 18. Take two answers and move on.
34	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | A near double is 1 more than the smaller double. Test it. / Make a puzzle | Write three near doubles for your grown-up / Teach it back | Show how to add 7 + 8 with a double / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your kit page and counters. Pause or rewind any time.*	Gather your workbook, pencil, paper, kit pieces and small things. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Show your answers on your fingers before I say them. Pause if you need longer.*	Play along. Show each answer on your fingers before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Build the doubles with red and yellow counters, then try one more and one less. Press play when you have tried.*	Pause. Build each double with red and yellow counters. Then add 1 more and take 1 away. Say each total.
27:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
39:00	Brain break	*Recording: stand up and join in! Stomp and clap each double with me.*	Stand up. Stomp the number on each foot and say the double.
43:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
71:00	Talk time	*Recording: pause. Tell a grown-up who is right, and show them with counters.*	Pause. Explain your answer to someone at home, with counters and a reason.
77:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
83:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
86:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:28	1:39	ok
remember	6:00	4:51	5:44	ok

sneak	4:00	3:04	3:35	ok
explore	15:00	12:34	14:55	ok
mini	12:00	10:07	11:57	ok
break	4:00	3:18	3:55	ok
work	28:00	22:59	27:25	ok
talk	6:00	4:54	5:50	ok
exit	6:00	4:58	5:53	ok
look	3:00	2:23	2:48	ok
flex	4:00	0:46	0:53	ok
Total	90:00	71:22	84:34	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 6.5 min for page 1 and 7.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.